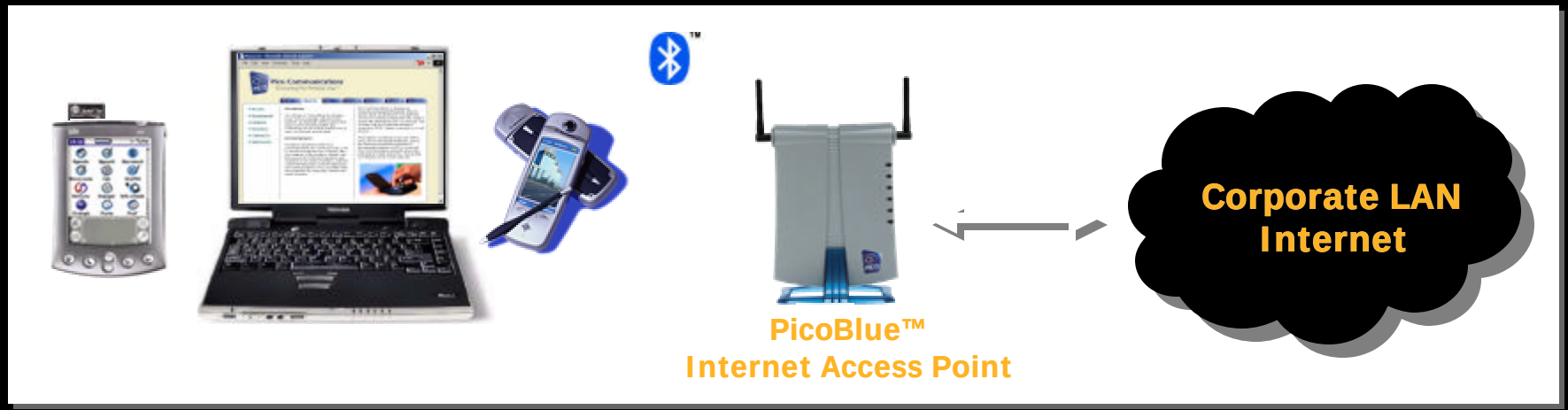


Developing Network Applications that Connect via a Bluetooth Access Point

Dave Bolan
VP Business Development
www.pico.net
davebolan@pico.net

Pico's Mission
"Connecting Our Wireless Lives"



Connecting portable computing devices
to the corporate Ethernet LAN and the Internet
via Bluetooth wireless technology
for the mobile professional
in the office, on the road, and at home

PicoBlue™ Internet Access Point





- Multipoint operation
  Seven simultaneous users
- Extended Range
  up to 100 m for Palm
  up to 250 m for PC
- DSL-like Speeds
  up to 723 kbps
- Bluetooth 1.1 Qualified

PicoOffice™ Client Software Suite



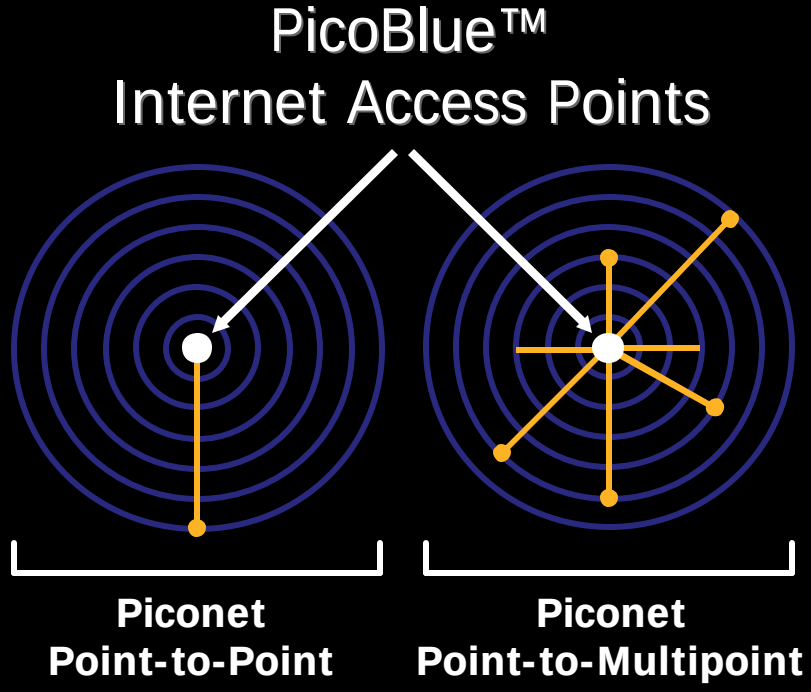
- PicoOffice™ brings new usage models to the Palm handheld computer
- A user can now be connected wirelessly through Bluetooth
- HotSync® operation is just “one-click away with PicoSync™
- No need to find a cradle or be close to your desktop



Topics

- What is Bluetooth
- Bluetooth in the Enterprise
- Wireless Networking Benefits
- Bluetooth and 802.11
- Meeting Enterprise Requirements
- Pico Communications Product Offerings
  - PicoBlue™ Internet Access Point
  - PicoConnect™
  - PicoManager™

Bluetooth – a networking technology



- Piconet
  - 1 to 7 simultaneous users
  - Up to 255 park devices
- Data Rate
  - Raw data
    - 1 Mbps
  - Asymmetric data
    - Up to 723.2/57.6 Kbps
  - Symmetric data
    - Up to 433.9 Kbps

Who is Behind Bluetooth?



- Bluetooth (SIG)

  Special Interest Group
  - Develop standards
  - Drive development
  - Promote the technology
  - Bluetooth Qualification Board (BQB)
    - Governs testing & qualification
  - 9 Promoter Companies
  - 187 Associate Companies (Pico)
  - 2640 Early Adopter Companies

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Bluetooth in the Enterprise

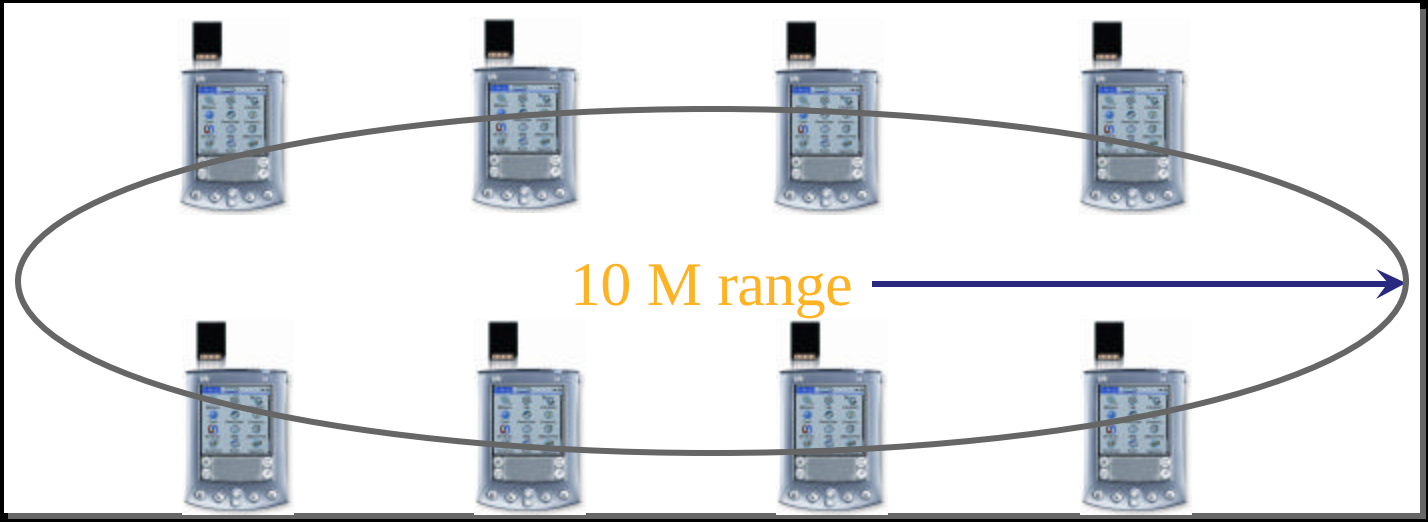
Mobile device connectivity, Ad-hoc networking,
Data access

Mobile Device Connectivity



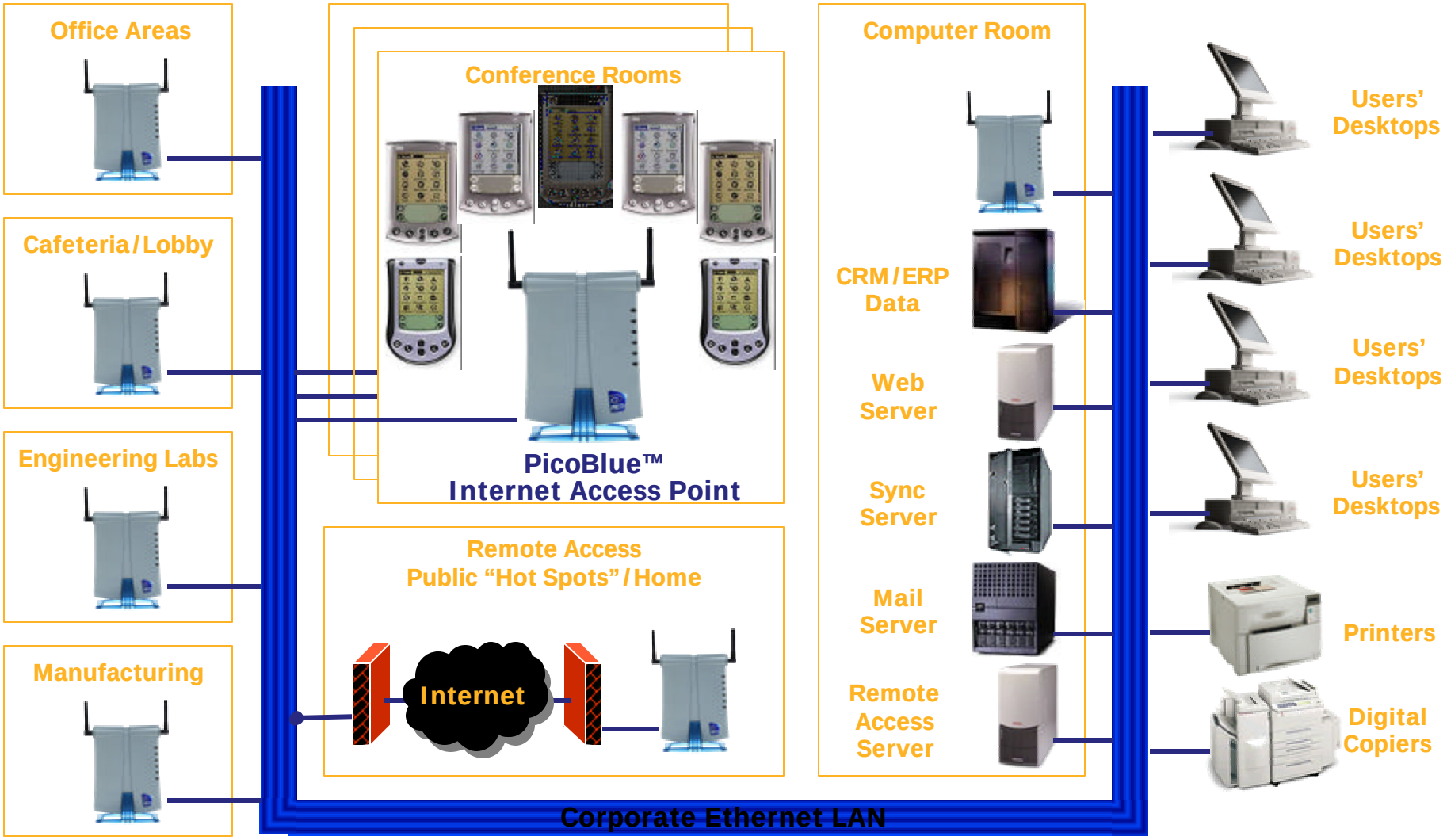
- Cordless headset
- Data access via mobile phone
- Printing
- Data synchronization

Ad-hoc Networking



- Temporary Personal Area Networks – PicoNets
  - Up to 8 members
  - No frequency planning required
- Peer-to-peer collaboration
  - Calendar synchronization
  - Chat
- Exchange vCards

Data Access to all Corporate Resources





Bluetooth Summary

- Bluetooth is **the** Personal Area Network standard
- Bluetooth is a perfect companion for mobile devices:
  - Small, low cost, minimal power
  - Plenty of throughput for most applications (723 kbps)
- Bluetooth has broad applicability:
  - Mobility and collaboration for notebook and PDA users in the enterprise
  - Inter-device communication, connection and data transfer in the home and office
  - Public access to the Internet via PDA and phone

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Wireless Networking Benefits

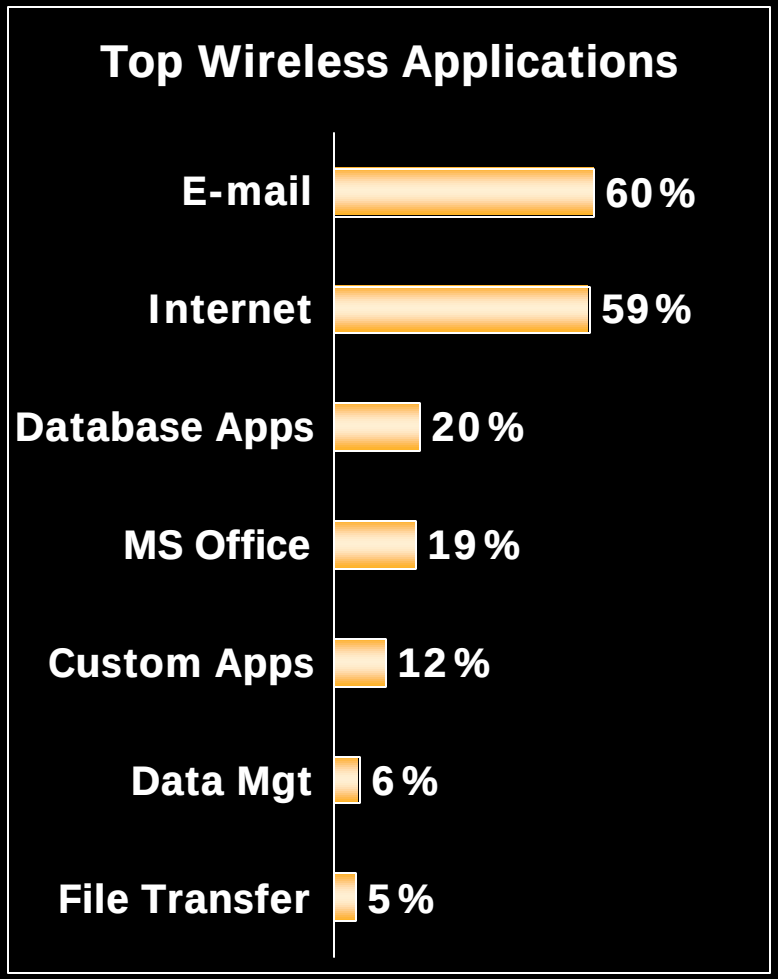


Wireless Networking Benefits

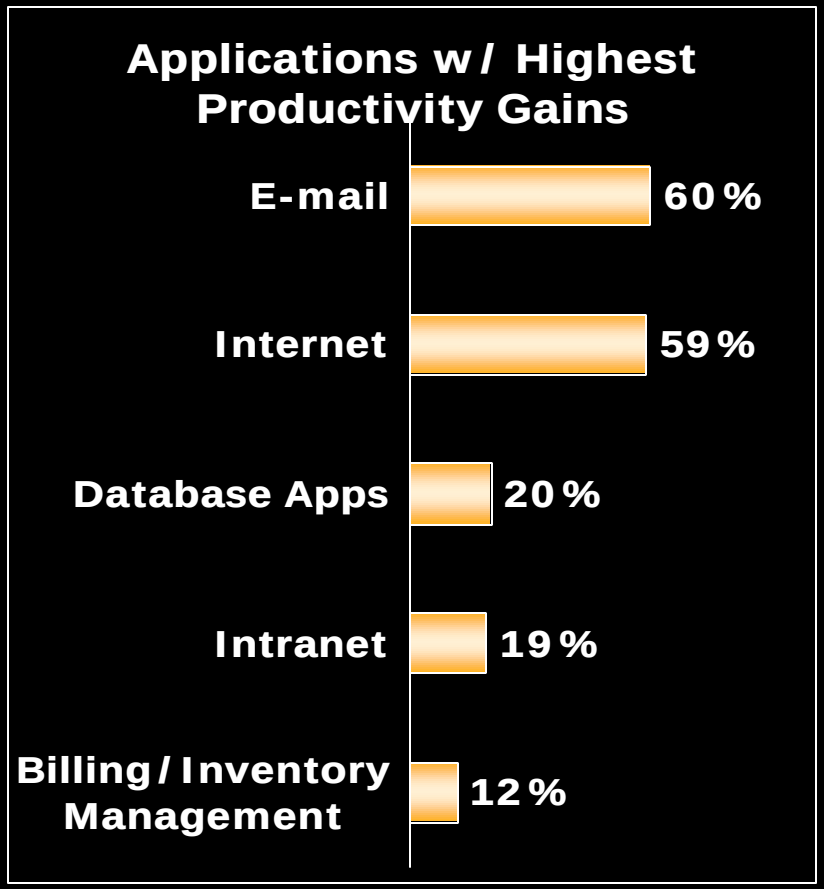
- 10% of US organizations have wireless networks
  - 16% of these employees have wireless access
- Each user is connected 1¾ hours more per day
  - 22% more productive
  - $7K annual productivity improvement
- ROI is $550 per user (without productivity gain analysis)
- 53% end-users have improved accuracy
  - 26% slight improvement
  - 27% significant improvement
  - 51% in healthcare have significantly increased accuracy
- 87% believe quality of life has improved

Source: NOP World – Technology, Fall 2001

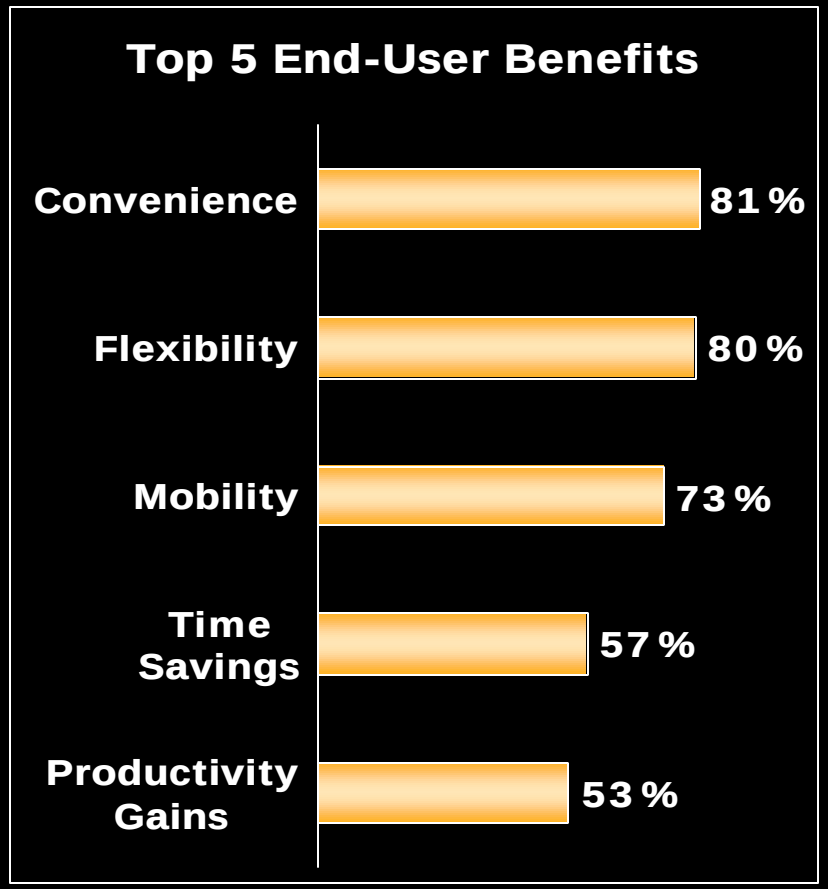
Applications in Wireless Networking

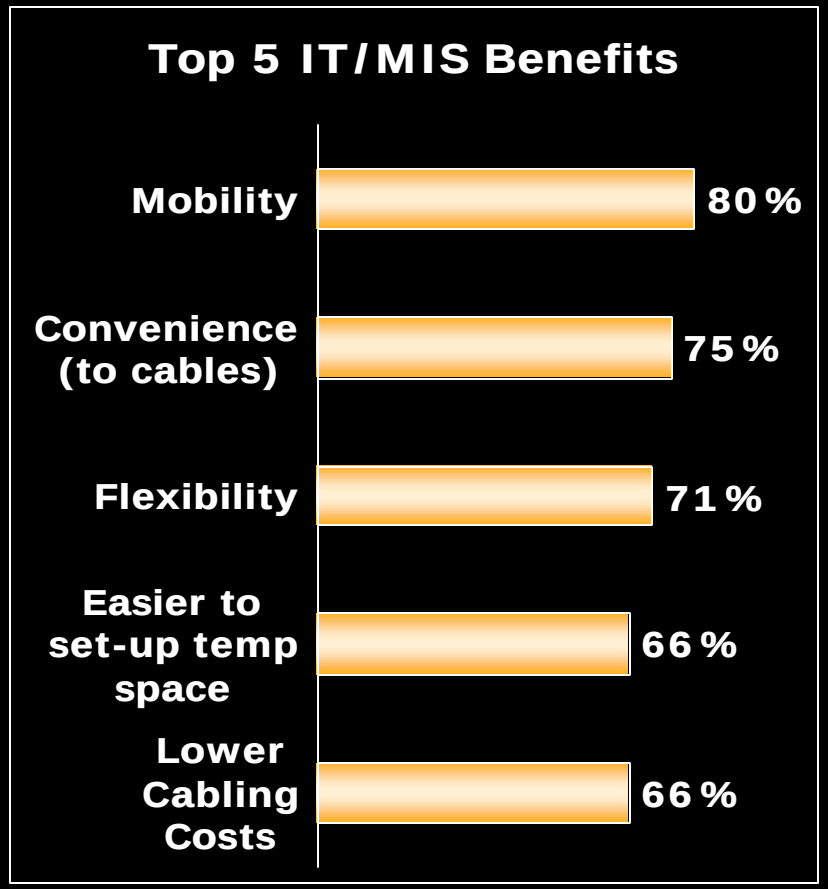


Source: NOP World – Technology, Fall 2001



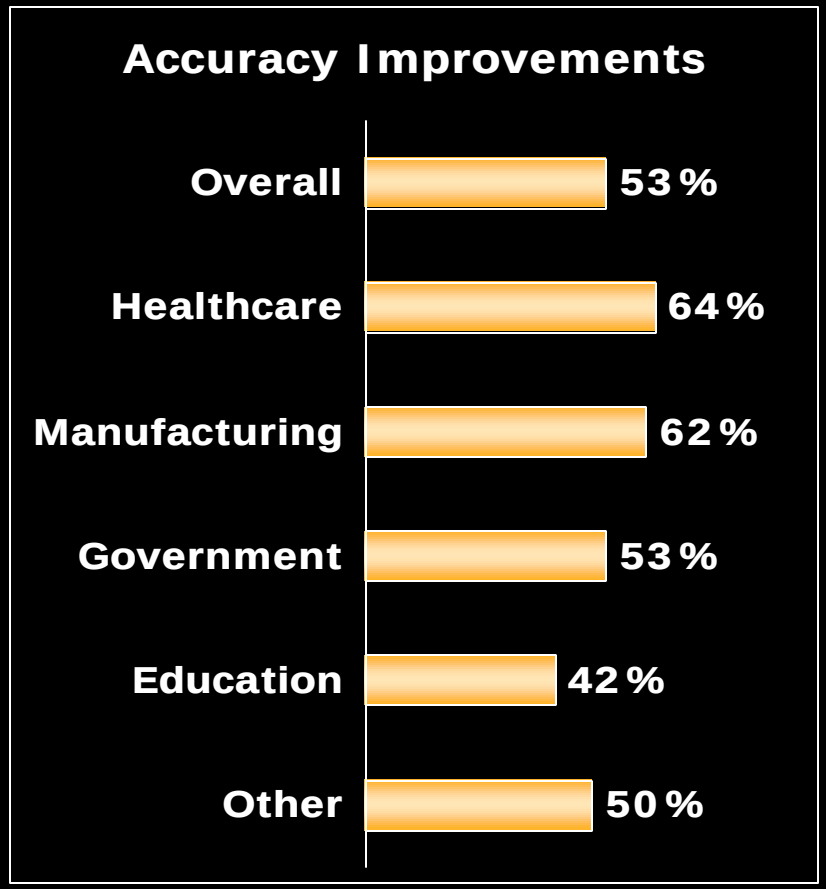
Benefits of Wireless Networking





Source: NOP World – Technology, Fall 2001

Benefits of Wireless Networking



Source: NOP World – Technology, Fall 2001



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Meeting Enterprise Requirements

Ease-of-Use, Security, Management, Co-existence



Ease of Use

- Deployment
  - PicoBlue provide greatest range
  - 7:1 connectivity enables greatest number of users
  - No separate subnet required
  - No frequency planning required
  - “Plug-n-Play” with DHCP
- Accessing administration functions over:
  - Bluetooth
  - Console port
  - LAN



Security

- Discoverability can be turned off
- Bluetooth Address filtering
- PPP authentication – PAP or CHAP MD5 protocol
- Service level or link level enforced security
- 128-bit private user-key authentication
- 8 to 128-bit private user-key encryption
- RADIUS and TACACS+ for AAA functions



Management

- Perform administration functions through:
  - HTTP web browser interface
  - TELNET command line interface
  - RS232 command line interface
  - PicoPing™ - access to all PicoBlue units
- DHCP client & server – Auto-configuration
- SNMP agent – centralized network monitoring
- RADIUS or TACACS+ – centralized AAA
- FTP client – remote firmware update/upgrade



Co-existence and Interference

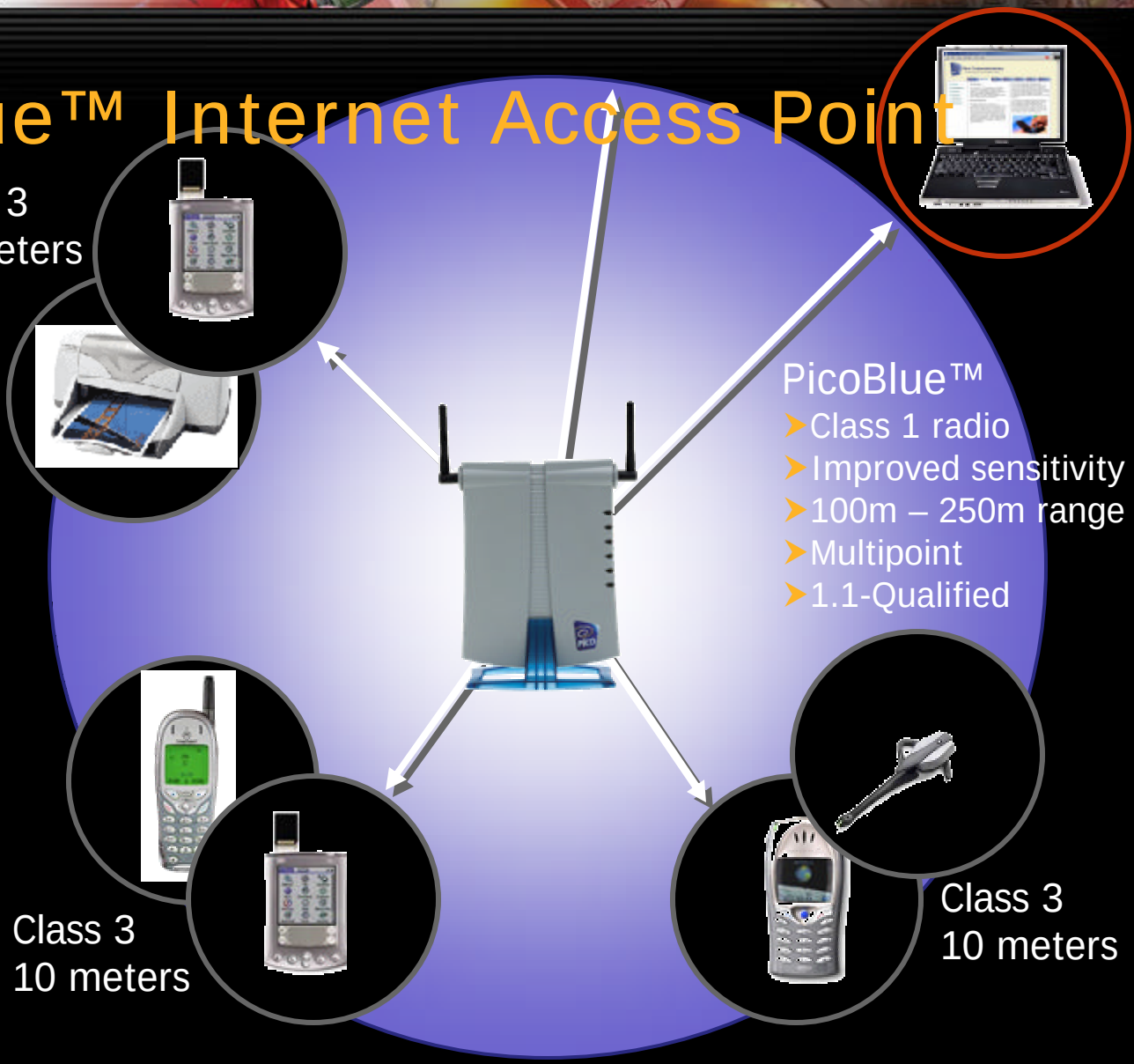
- Bluetooth and 802.11b
  - Both operate in 2.4 GHz ISM band
  - Interference with <10 cm separation
  - Co-exist with >1 meter separation
- Bluetooth and 802.11a
  - Operates in different bands
    - Bluetooth: 2.4 GHz
    - 802.11a: 5.2 GHz and
  - No measurable RF interference expected



Pico Product Offerings

PicoBlue Internet Access Point, PicoConnect™, PicoManager™

PicoBlue™ Internet Access Point

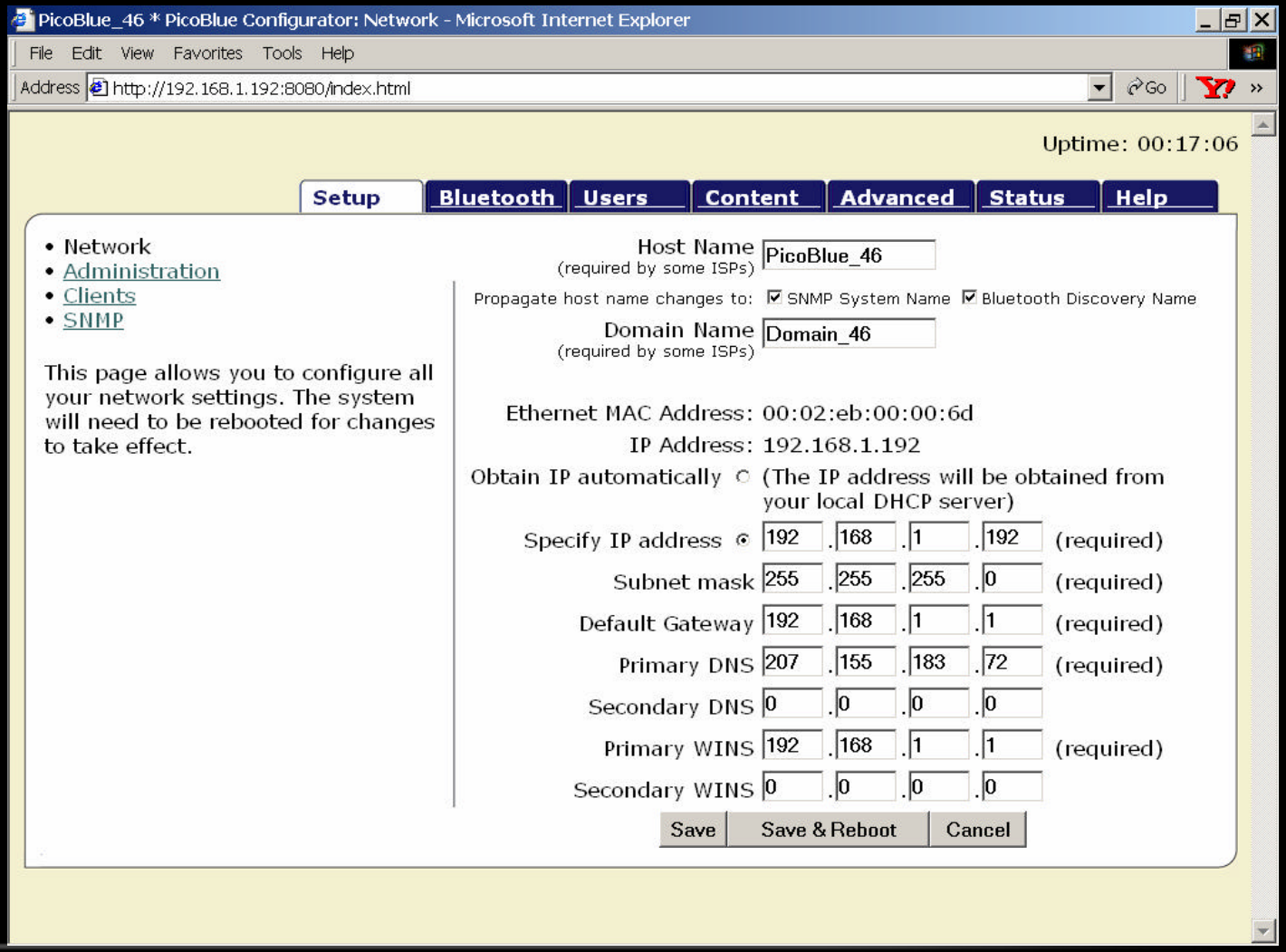


PicoOffice™ Client Enterprise Features

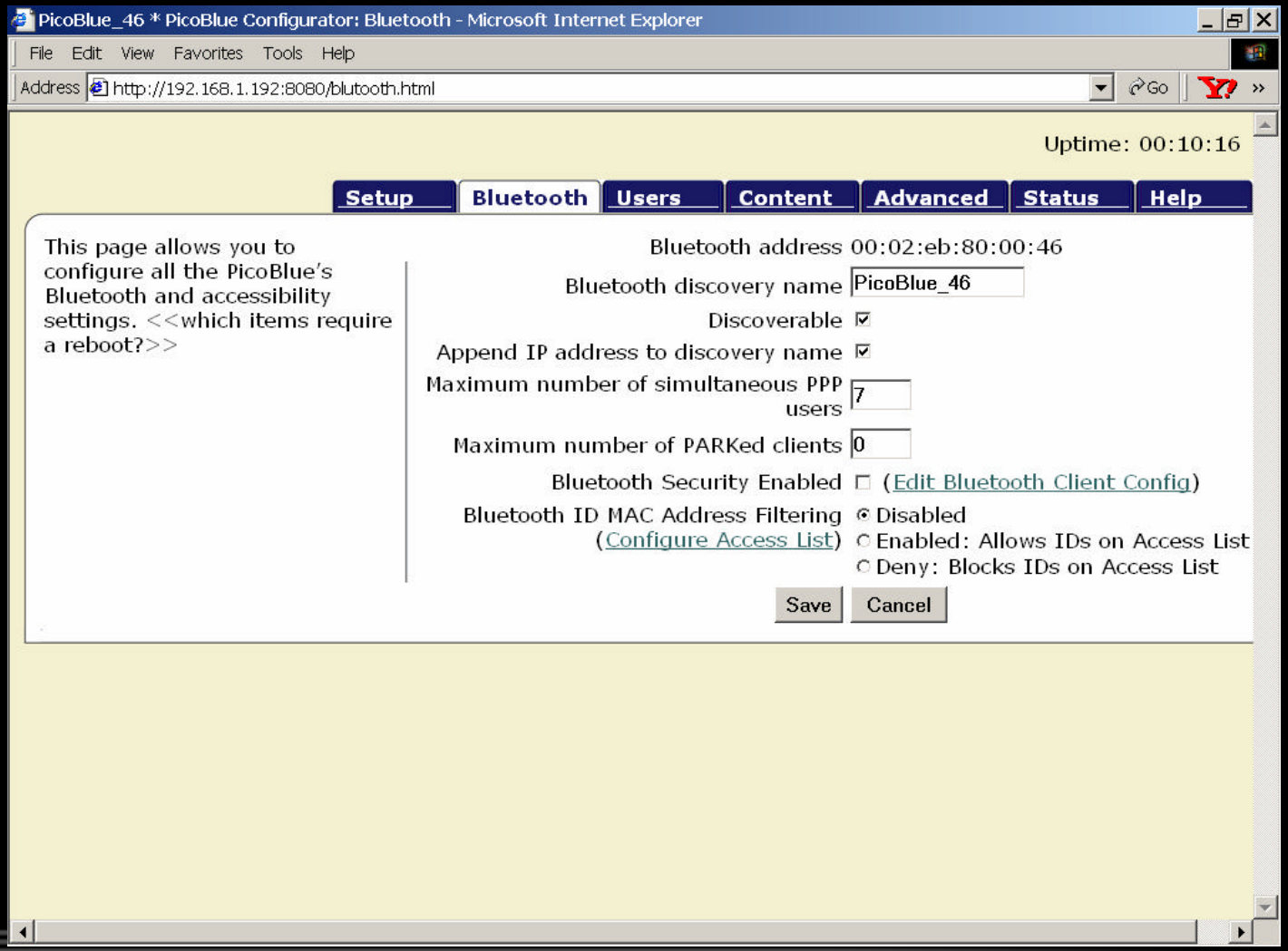


- PicoConnect™
  - Automates Bluetooth network access
  - Finds all PicoBlue™ AP
  - Checks availability of APs
  - Selects optimal AP
    - Performs load balancing between AP
  - If no AP available
    - **Tries Bluetooth cell phone**
    - **Always connected**
- PicoSync™
  - Special version of PicoConnect
  - Performs Wireless Network HotSync® Operation

PicoManager™ Set up



PicoManager™ Set up



Covering the Bluetooth Congress 2001

Bluetooth daily news

Thursday June 7 2001

www.thebluelink.com

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See pages 10-11 & 20

P.8
A VIEW FROM THE FRONTLINE
So where are the millions of chips being used and what issues stand in the way of growing this number?

P.13
OPTIMISED FOR SUCCESS
Will we soon be seeing a family of Bluetooth chips

PICO HAS 1.1 QUALIFICATION



Pico Communications has received an eleventh-hour nod from the Governor informing the company that its point-to-multipoint Bluetooth LAN access technology wasn't just standard 1.1 compliant: it was Qualified. Press releases were duly crossed out and redrafted in bins.

Pico reckons that makes them the only access point producer but one to be qualified for standard 1.1, and the only one to offer multiple channels of access through one access point. The new standard specifies for 7 live sessions, and 256 "parks", idle users who can remain logged in, but who have to wait for an active user to stop before they can transmit data.

The access points have a range of up to 100 metres, according to Devin Bolan, Pico's VP of business development, he can walk the four corners of the exhibition floor and stay connected to the access point at his stand. Security for the access points can be maintained either through remote authentication, by establishing a list of accepted users, or by disabling the mechanism by which the access points respond to searching Bluetooth devices.

Pico seems to be of the opinion that it will be the business users, whether that be in Enterprise or small offices, that will drive the initial pick-up of Bluetooth. However, they also anticipate a substantial market in public spaces such as airports and supermarkets. But while a 100 metre range is pretty good, the lack of Bluetooth hand-offs might keep widespread public use to a minimum for the time being.

Product Summary

PicoConnect™ Client Solutions



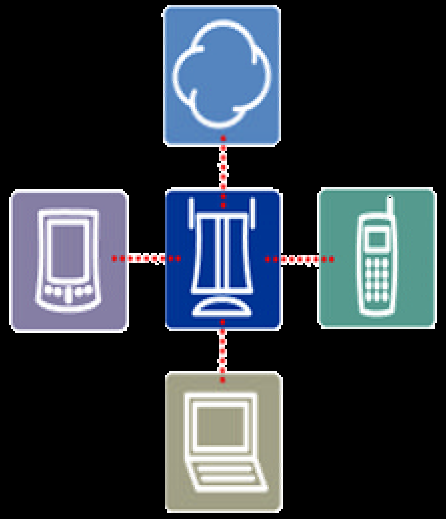
- Auto network access
- Load balancing
- Selects optimal AP
- Defaults to cell phone
- PicoSync™

PicoBlue™ Internet Access Point



- Extended range
- Point-to-multipoint
- Demo. interoperability
- Complete networking
- Enterprise features

PicoManager™ Server Suite



- Enable large scale deployments
- Standards-based management
- Access control and accounting
- URL redirect
- PicoPing™


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